GEE Lab - 1

WORKING WITH GOOGLE ERATH ENGINE

TO SEARCH, FIND AND VISUALIZE A BROAD RANGE OF REMOTELY SENSED DATASETS

Free cloud processing with built-in functions

GOOGLE EARTH ENGINE

A CLOUD-BASED PLATFORM FOR PLANETARY-SCALE GEOSPATIAL ANALYSIS

No more downloading and analyzing individuals tiles at a time

Sign up for a Google Earth Engine account

Earth Engine account registration: https://signup.earthengine.google.com/#!/

Once registered you can access the Earth Engine environment @:
https://code.earthengine.google.com

Google Earth Engine Code Editor uses the JavaScript programming languages.

For JavaScript Background: https://developers.google.com/earth-engine/tutorial_js_01

Google Mission Statement

"To organize the world's information and make it universally accessible and useful."

Earth Engine provides an API to process, analyze, and visualize the data using Google machines.

Data storage and processing are all performed at Google.

Whatever USGS has, Earth Engine has.

All you need to use Earth Engine is an internet connection.

Earth Engine Data Catalog

- No more downloading and analyzing individuals tiles at a time.
- No more worry about your local storage.

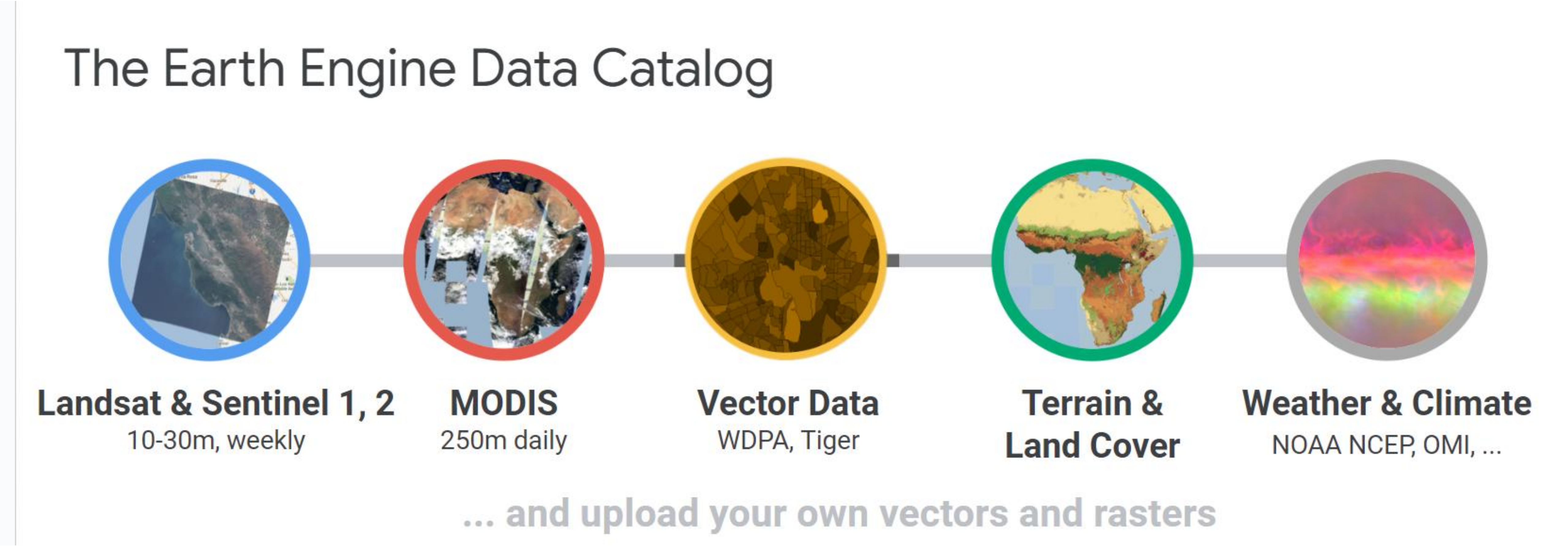
As of 2020

> 200 public datasets
> 5 million images
> 4000 new images every day
~ 17 petabytes of data

Now

- 1000 of public datasets
- Thousands of new images added every day
- Total Stored Data: Over 90 petabytes (PB) and expanding rapidly.
- Growth Rate: Grows by roughly 1 petabyte of new data every month.
- Historical Span: Over 50 years of continuous global satellite records.

https://developers.google.com/earth-engine/datasets/catalog/?filter=SRTM



There is a one or two-day latency between scene acquisition and when it is ingested into the catalog.

33 years

Of satellite data

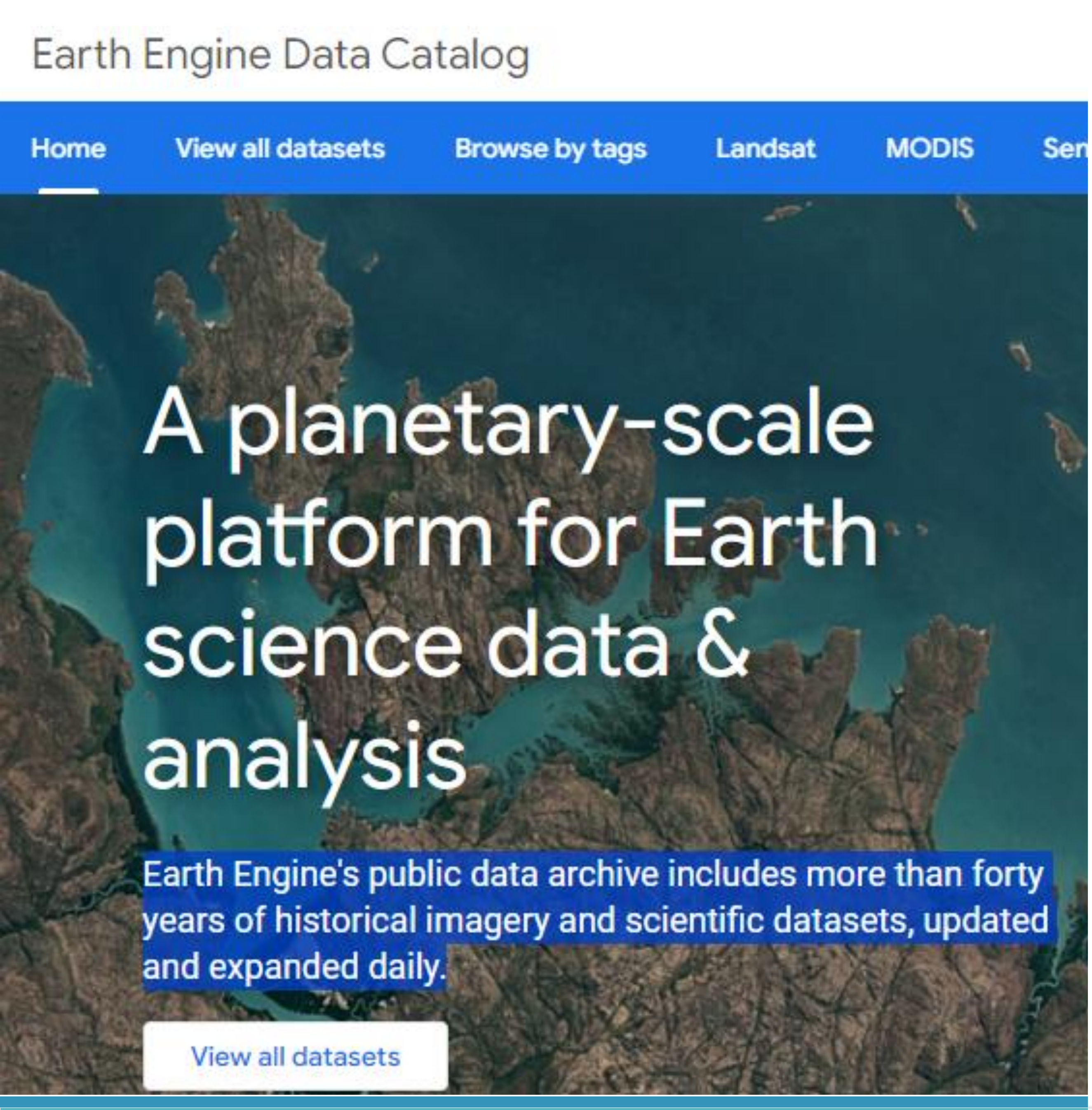
Over 5,000,000

Landsat and Sentinel scenes analyzed

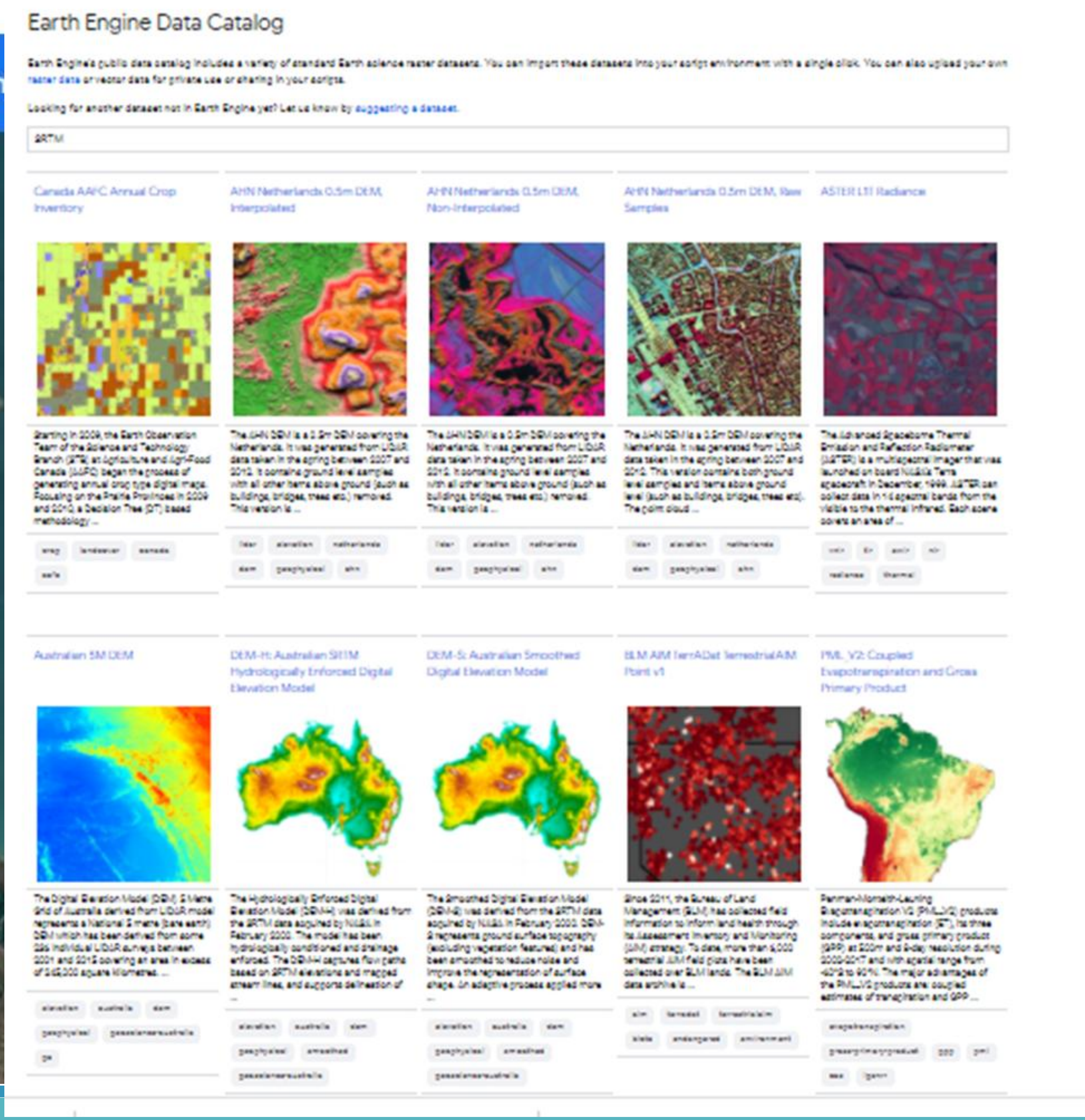
3 Quadrillion Pixels

3,000,000,000,000,000

- ✓ Originally launched more than a decade ago for research; now available for commercial use
- ✓ Ability to integrate with powerful analytics and tools like BigQuery and Vertex AI
- ✓ Remains at no cost for nonprofit, noncommercial, and research projects; learn more

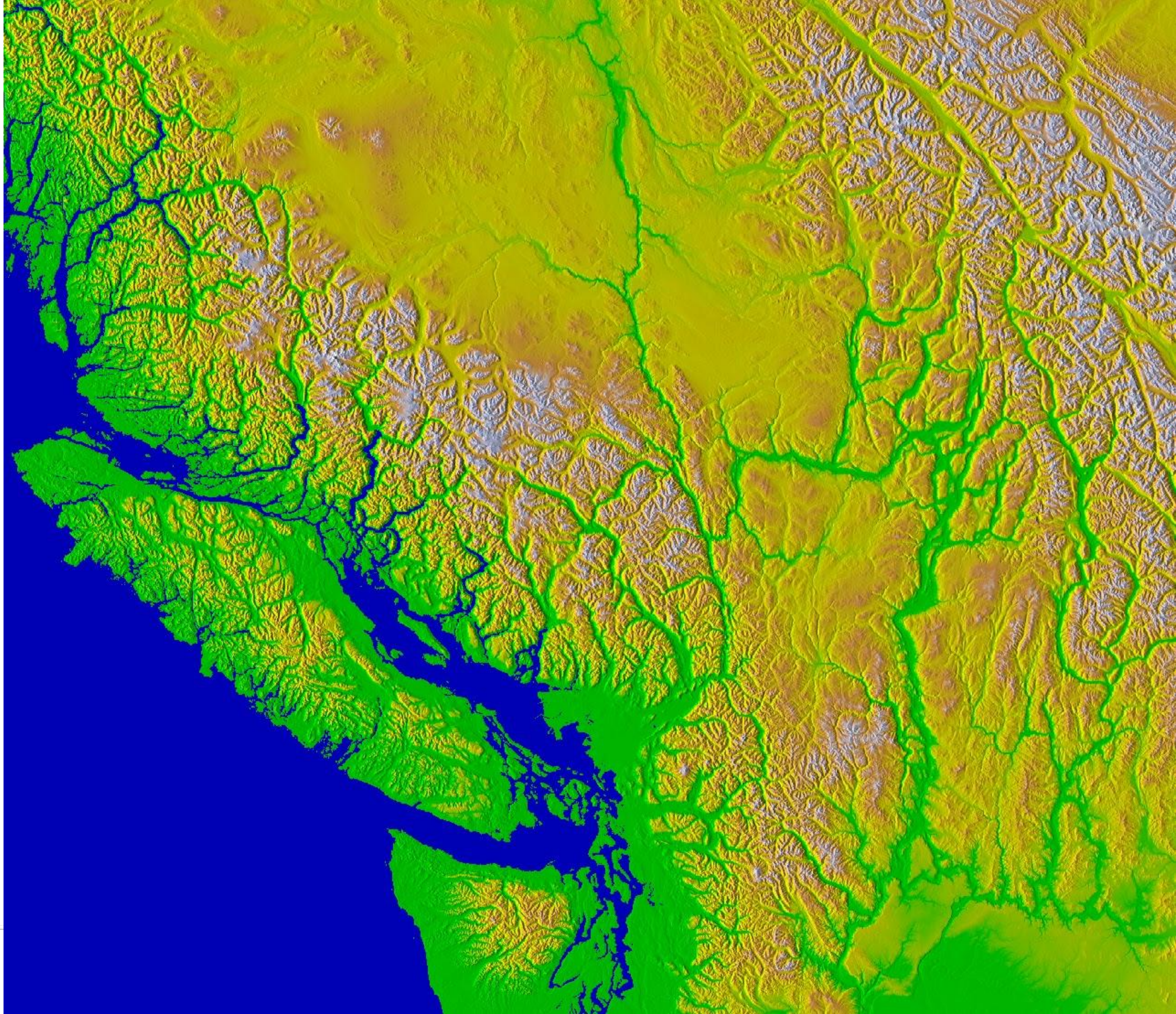


Earth Engine Data Catalog
Home
View all datasets
Browse by tags
Landsat
MODIS
A planetary-scale platform for Earth science data & analysis
Earth Engine's public data archive includes more than forty years of historical imagery and scientific datasets, updated and expanded daily.
View all datasets

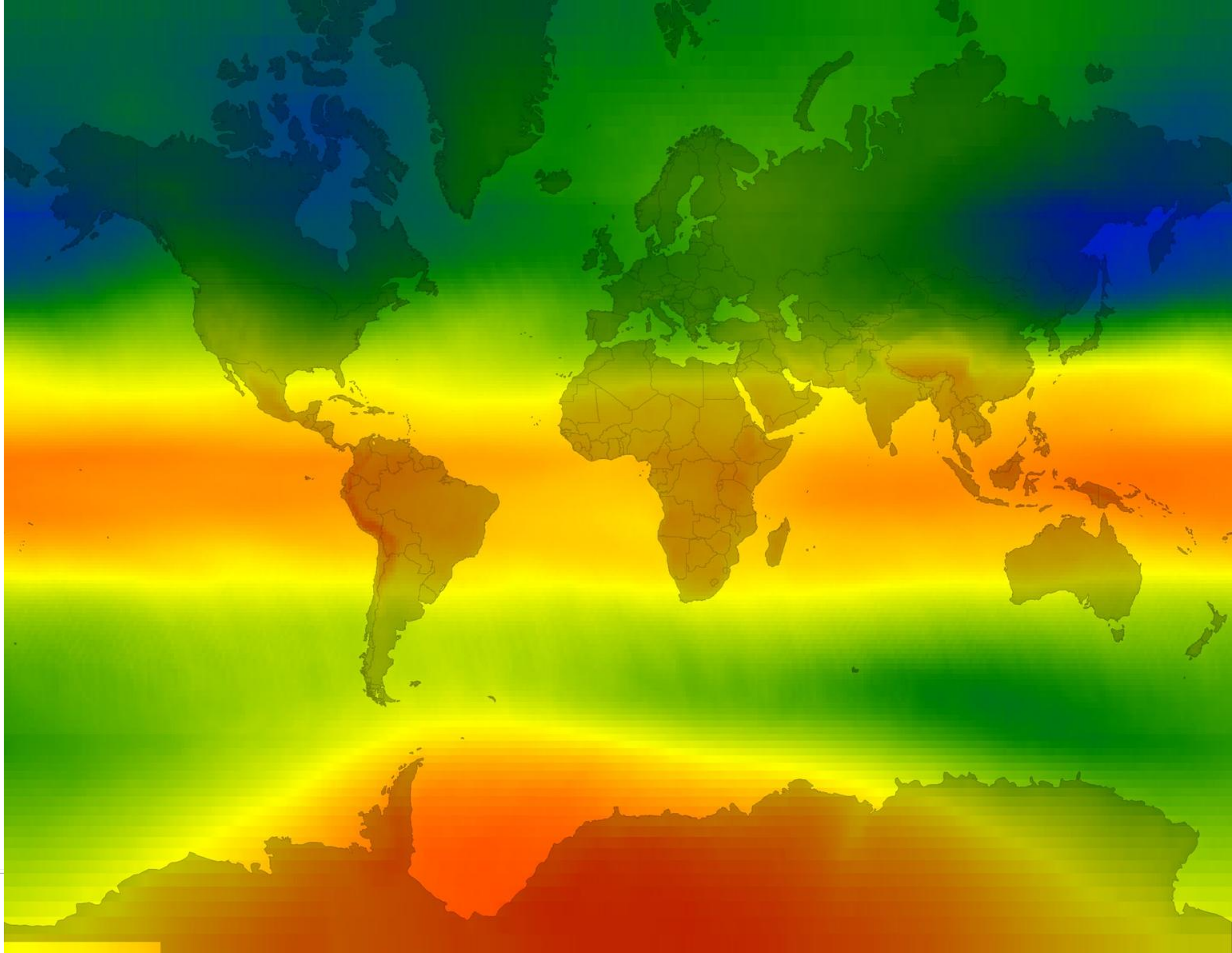


Earth Engine Data Catalog
Canada AAFC Annual Crop Inventory
AHN Netherlands 0.5m DEM, Interpolated
AHN Netherlands 0.5m DEM, Non-Interpolated
AHN Netherlands 0.5m DEM, Raw Samples
ASTER L1T Radiance
Australian 5M DEM
DEM-H: Australian SRTM Hydrologically Enforced Digital Elevation Model
DEM-S: Australian Smoothed Digital Elevation Model
BLM AIM TerrADat TerrestrialAIM Point v1
PML_V2: Coupled Evapotranspiration and Gross Primary Product

Terrain

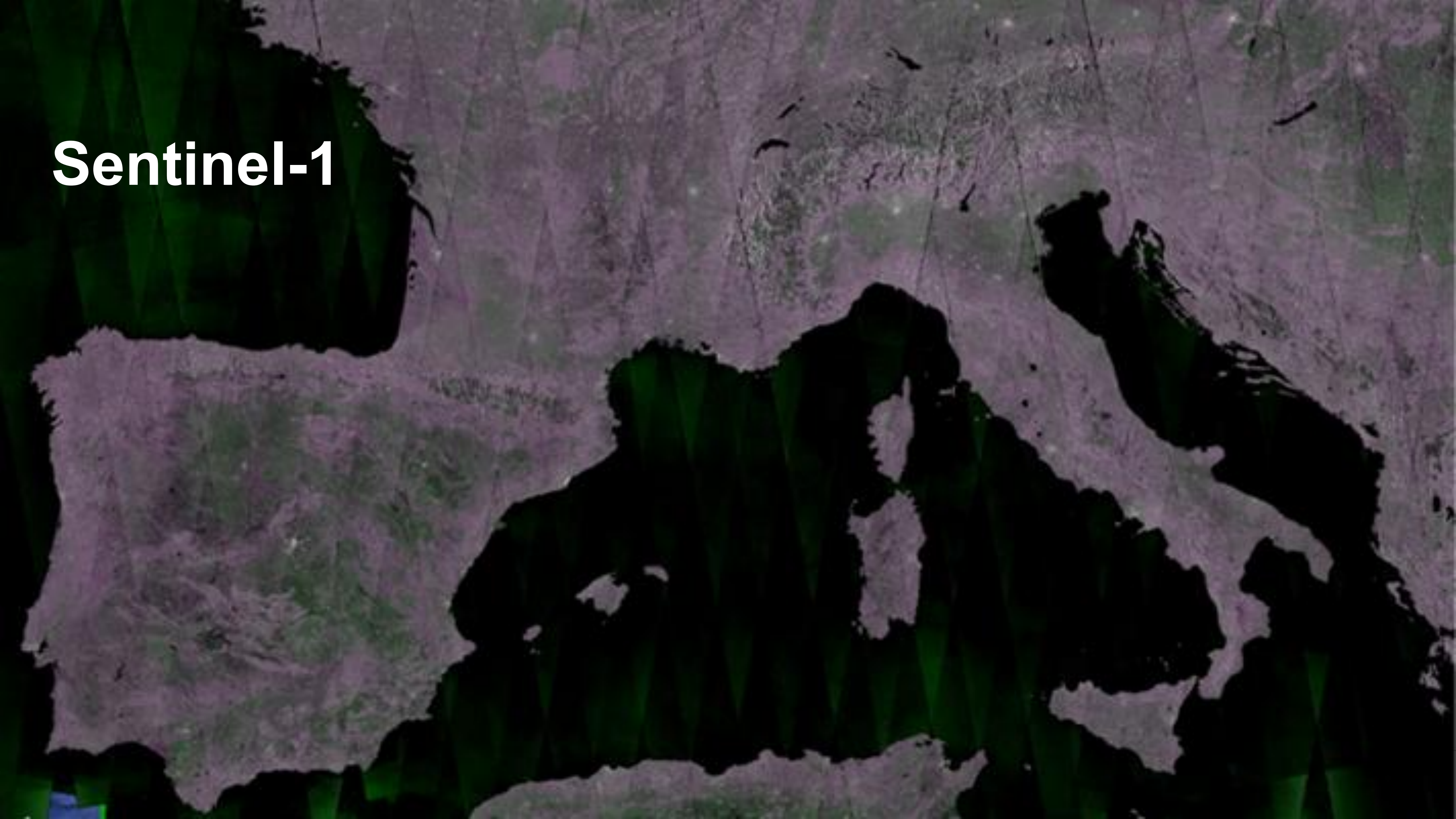


Atmosphere



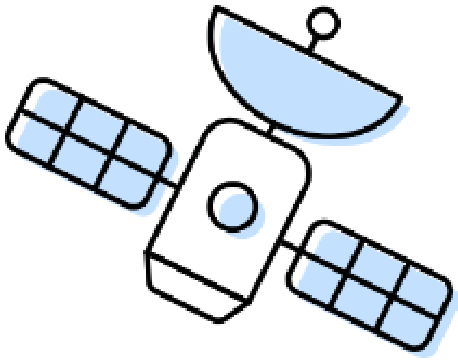
Google Earth Engine

Sentinel-1



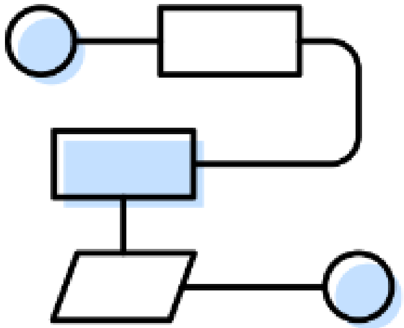
Meet Earth Engine

Google Earth Engine combines a multi-petabyte catalog of satellite imagery and geospatial datasets with planetary-scale analysis capabilities. Scientists, researchers, and developers use Earth Engine to detect changes, map trends, and quantify differences on the Earth's surface. Earth Engine is now available for commercial use, and remains free for academic and research use.



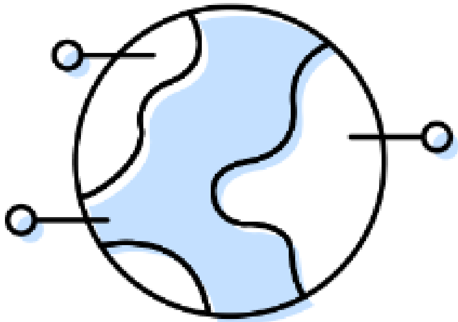
Satellite Imagery

+



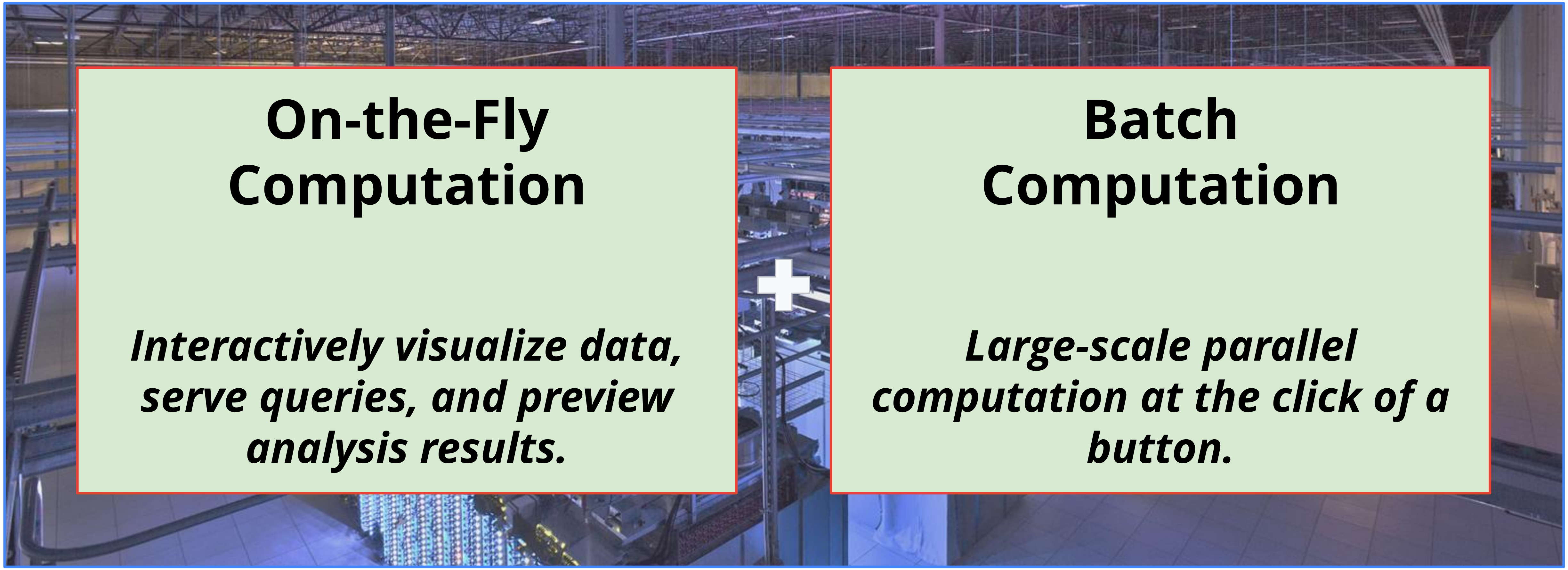
Your Algorithms

+



Real World Applications

Computation Platform



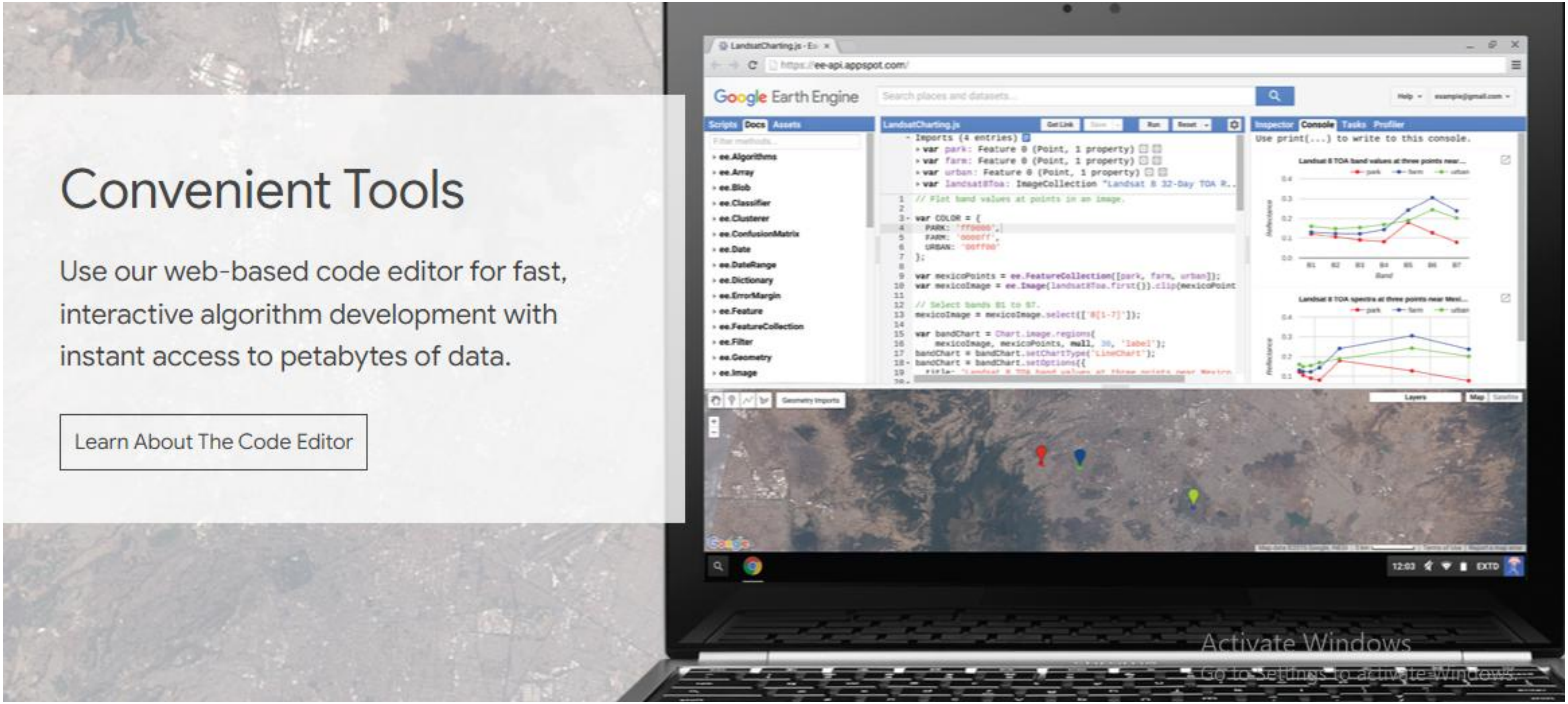
On-the-Fly Computation

Interactively visualize data, serve queries, and preview analysis results.

Batch Computation

Large-scale parallel computation at the click of a button.

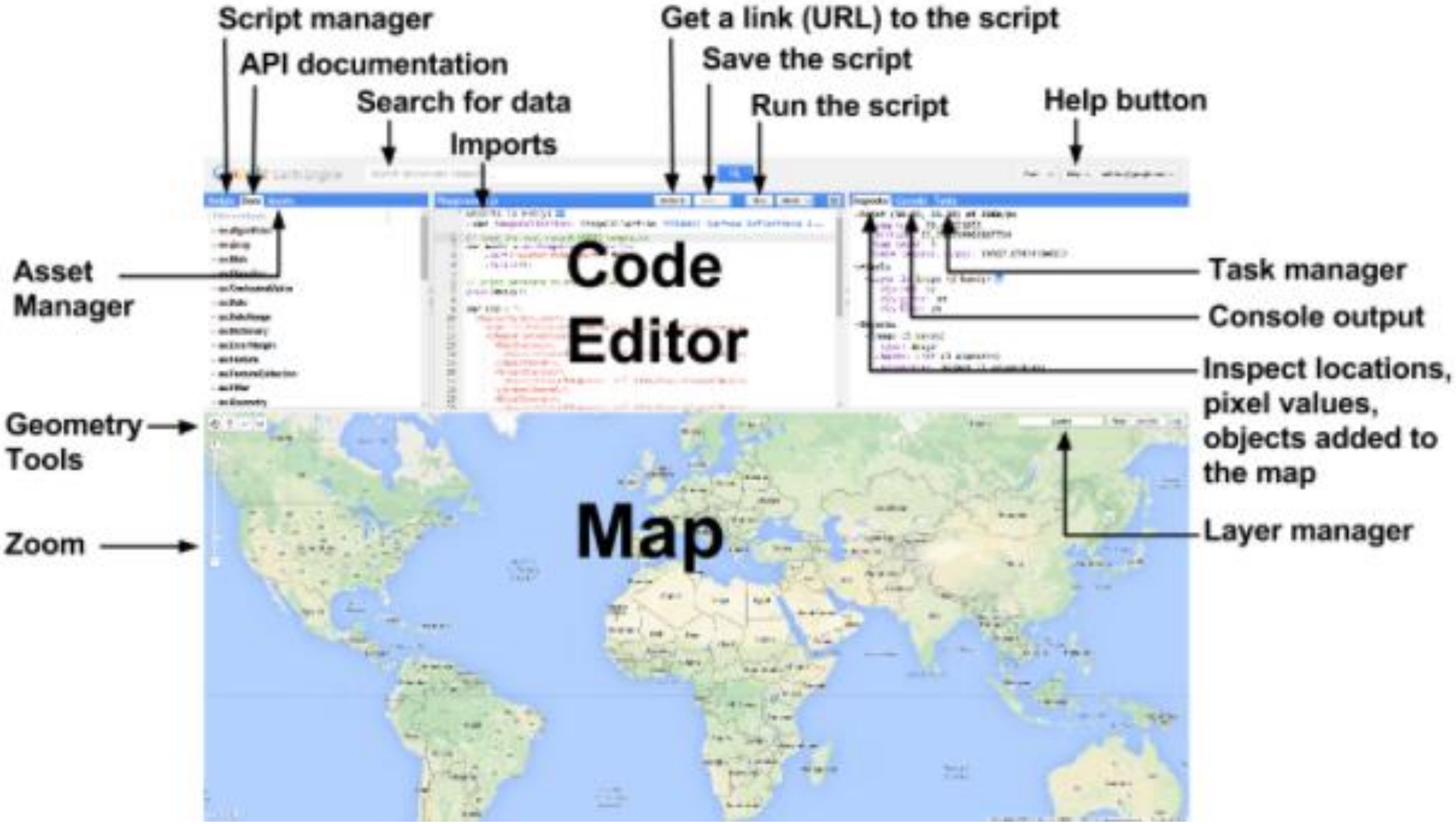
Powerful JavaScript* API

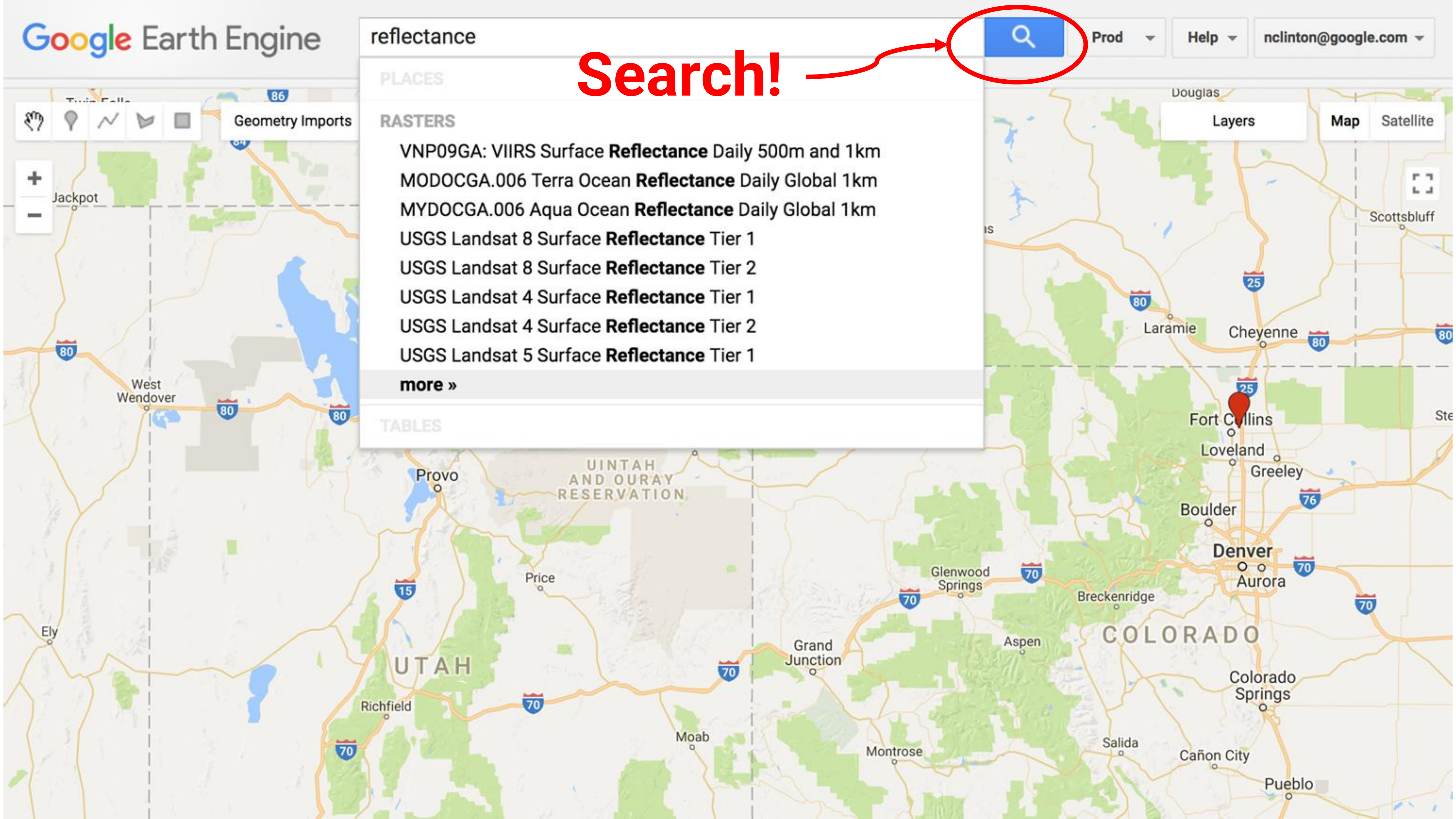


Javascript API

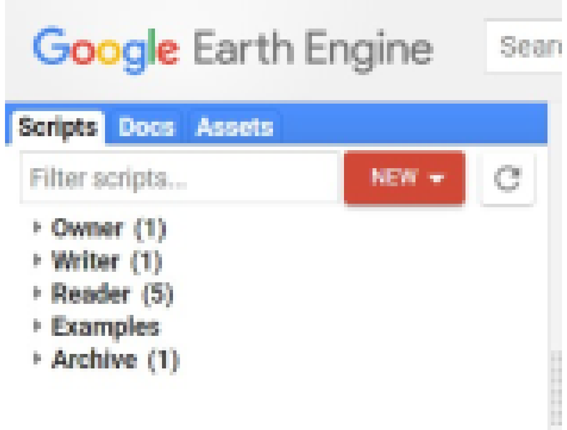
The Javascript API is a set of functions that allows you to interact with the data through a dedicated code editor.

1. The Earth Engine code editor




Google Earth Engine
reflectance
Search!
Prod
Help
nclinton@google.com
PLACES
RASTERS
VNP09GA: VIIRS Surface **Reflectance** Daily 500m and 1km
MODOCGA.006 Terra Ocean **Reflectance** Daily Global 1km
MYDOCGA.006 Aqua Ocean **Reflectance** Daily Global 1km
USGS Landsat 8 Surface **Reflectance** Tier 1
USGS Landsat 8 Surface **Reflectance** Tier 2
USGS Landsat 4 Surface **Reflectance** Tier 1
USGS Landsat 4 Surface **Reflectance** Tier 2
USGS Landsat 5 Surface **Reflectance** Tier 1
more »
TABLES
Geometry Imports
Layers
Map
Satellite
Jackpot
West Wendover
Provo
UINTAH AND OURAY RESERVATION
Price
UTAH
Richfield
Moab
Grand Junction
Montrose
Ely
Glenwood Springs
Aspen
Breckenridge
COLORADO
Salida
Cañon City
Pueblo
Colorado Springs
Denver
Aurora
Boulder
Greeley
Loveland
Fort Collins
Laramie
Cheyenne
Scottsbluff
Douglas

1

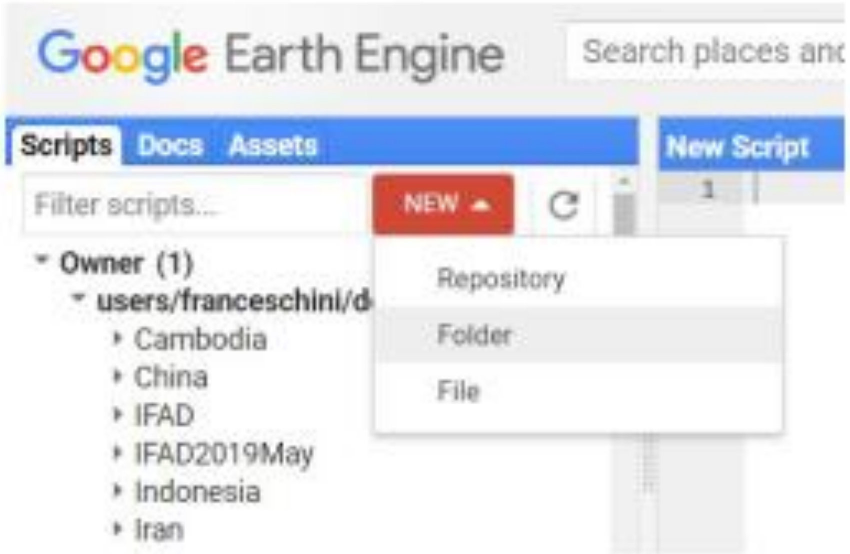


EXERCISE: CREATE A FOLDER

We will create a folder to store the scripts of the training:

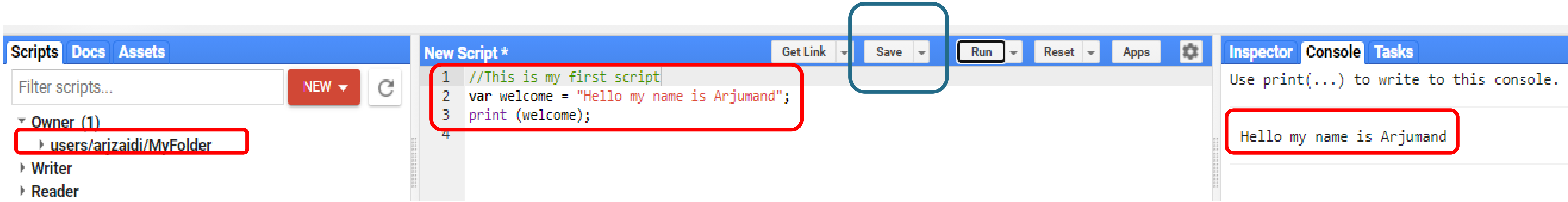
- Click on the NEW button
- Select Folder

2



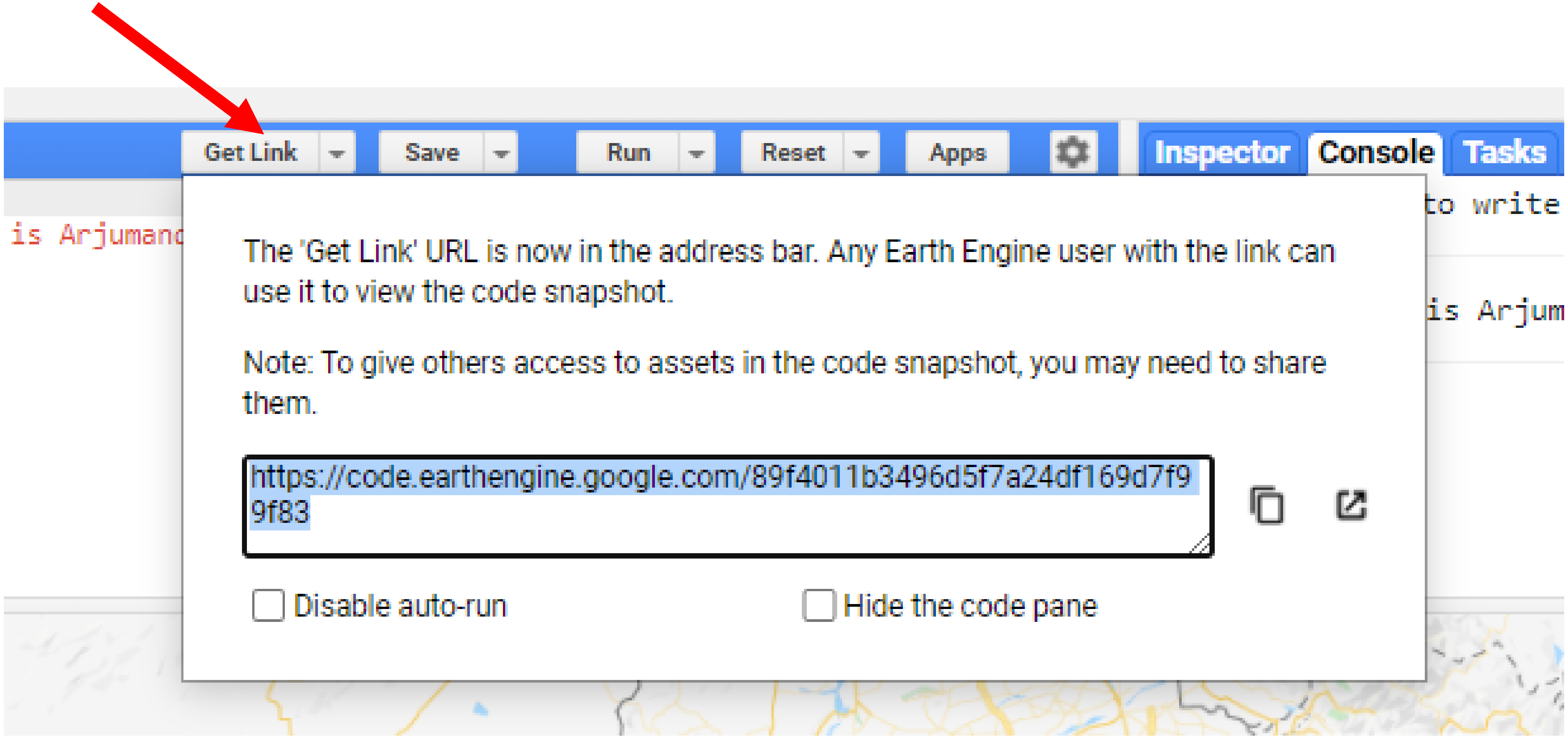
3 – Type name of the folder

Your first Script: Hello, I am 'your name'

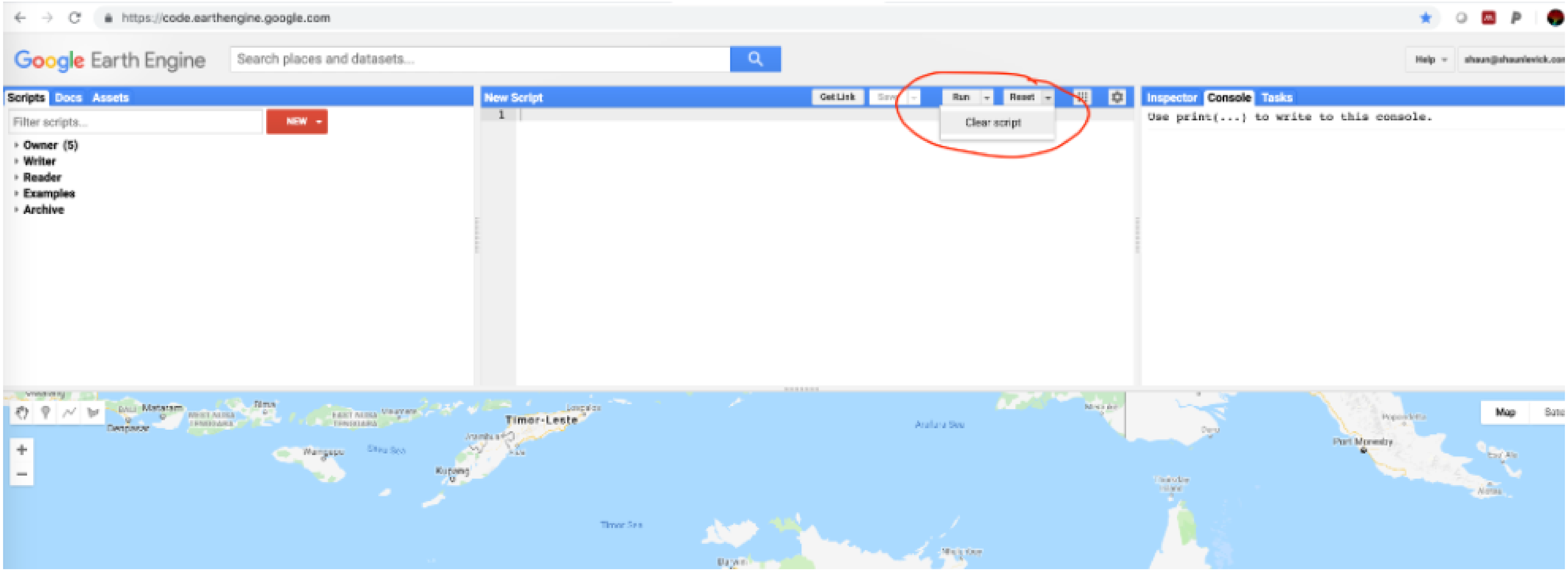


- Each variable has to be declared with the **var** operator
- Each line has to end with a semicolon
- Outputs are written in the Console panel
- Double slash is used to comment

The **Get Link** button, allows to generate an URL link of the script that can be shared with other users



Clear Script



Data Types and Geospatial Processing Functions

- **Image** - band math, clip, convolution, neighborhood, selection ...
- **Image Collection** - map, aggregate, filter, mosaic, sort ...
- **Feature** - buffer, centroid, intersection, union, transform ...
- **Feature Collection** - aggregate, filter, flatten, merge, sort …
- **Filter** - by bounds, within distance, date, day-of-year, metadata ...
- **Reducer** - mean, linearRegression, percentile, histogram ….
- **Join** - simple, inner, outer, inverted ...
- **Kernel** - square, circle, gaussian, sobel, kirsch …
- **Machine Learning -** CART, random forests, bayes, SVM, kmeans, cobweb …
- **Projection** - transform, translate, scale …

over 1000 data types and operators, and growing!

Global composites with a few lines of code

```
var composite = ee.Algorithms.Landsat.simpleComposite({
  collection: ee.ImageCollection('LANDSAT/LC08/C01/T1'),
  asFloat: true
});

Map.addLayer(composite,
            {bands: ['B4', 'B3', 'B2'], max: 0.3},
            'composite');
```

https://code.earthengine.google.com/05d2e23206b329dfe696e5ba8e232c3f

Contents lists available at ScienceDirect

Remote Sensing of Environment

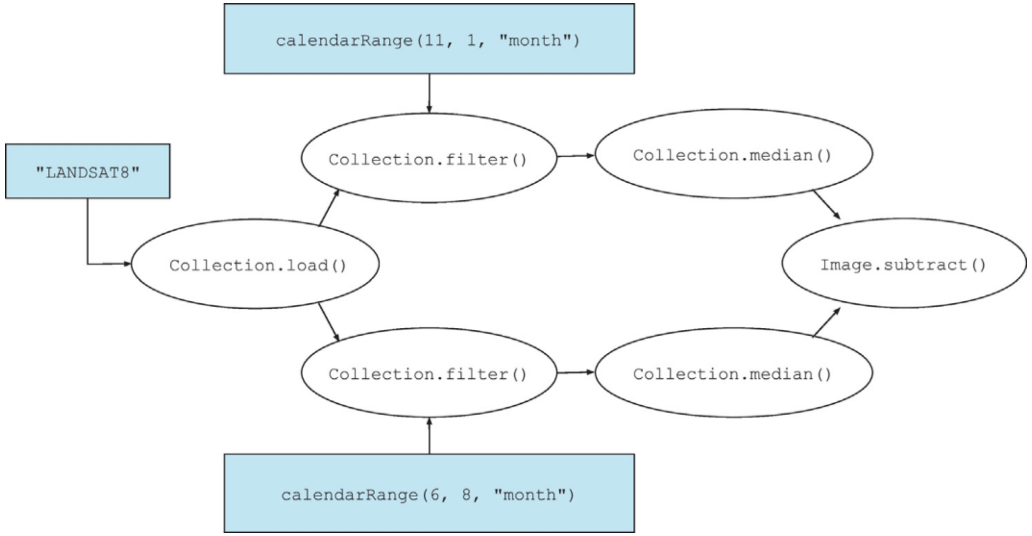
journal homepage: www.elsevier.com/locate/rse

Google Earth Engine: Planetary-scale geospatial analysis for everyone

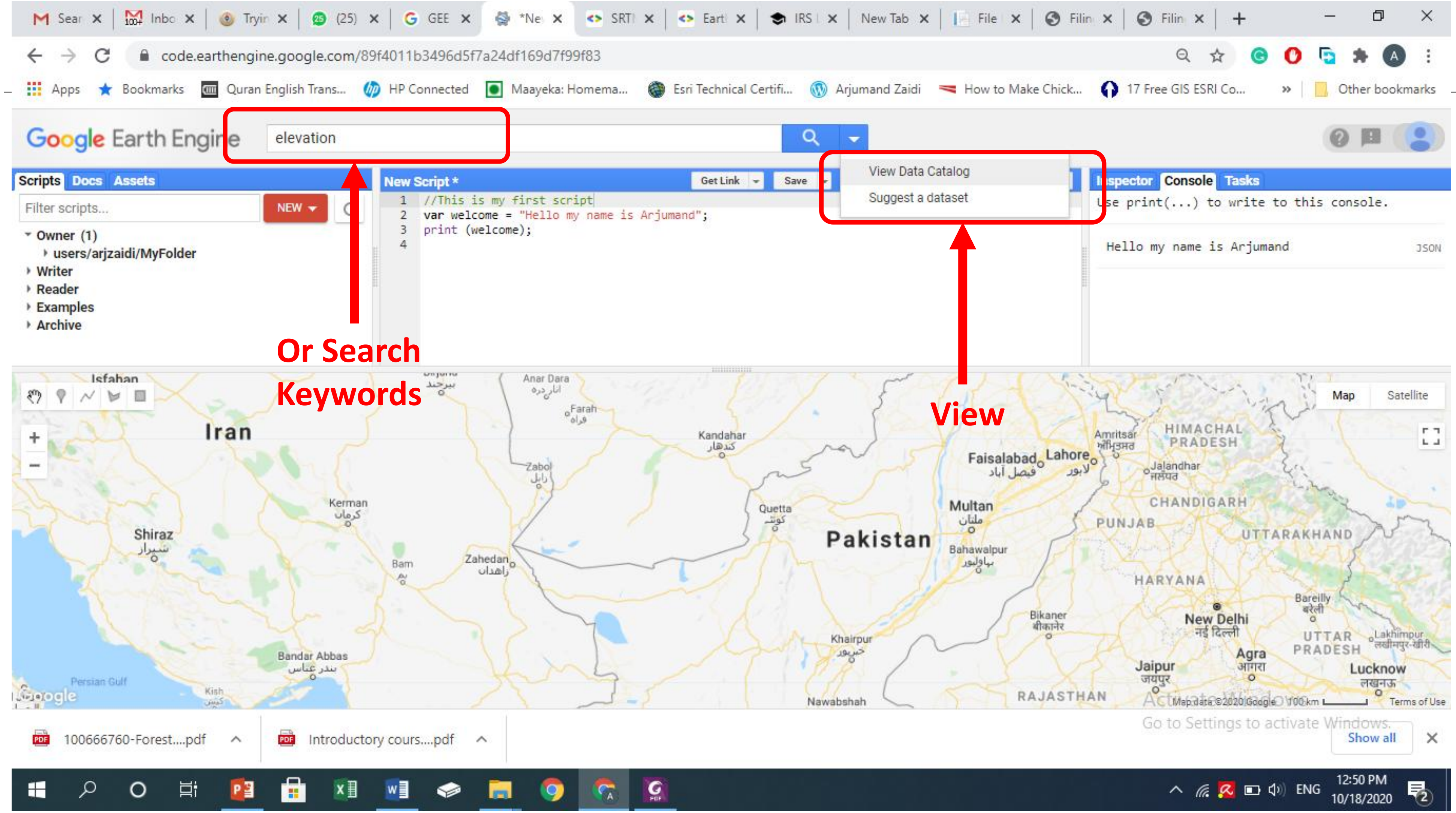
Noel Gorelick [a,*], Matt Hancher [b], Mike Dixon [b], Simon Ilyushchenko [b], David Thau [b], Rebecca Moore [b]

https://www.sciencedirect.com/science/article/pii/S0034425717302900

```
collection = ee.ImageCollection("LANDSAT8")
winter = collection.filter(ee.Filter.calendarRange(11, 1, "month"))
summer = collection.filter(ee.Filter.calendarRange(6, 8, "month"))
diff = summer.median().subtract(winter.median())
```



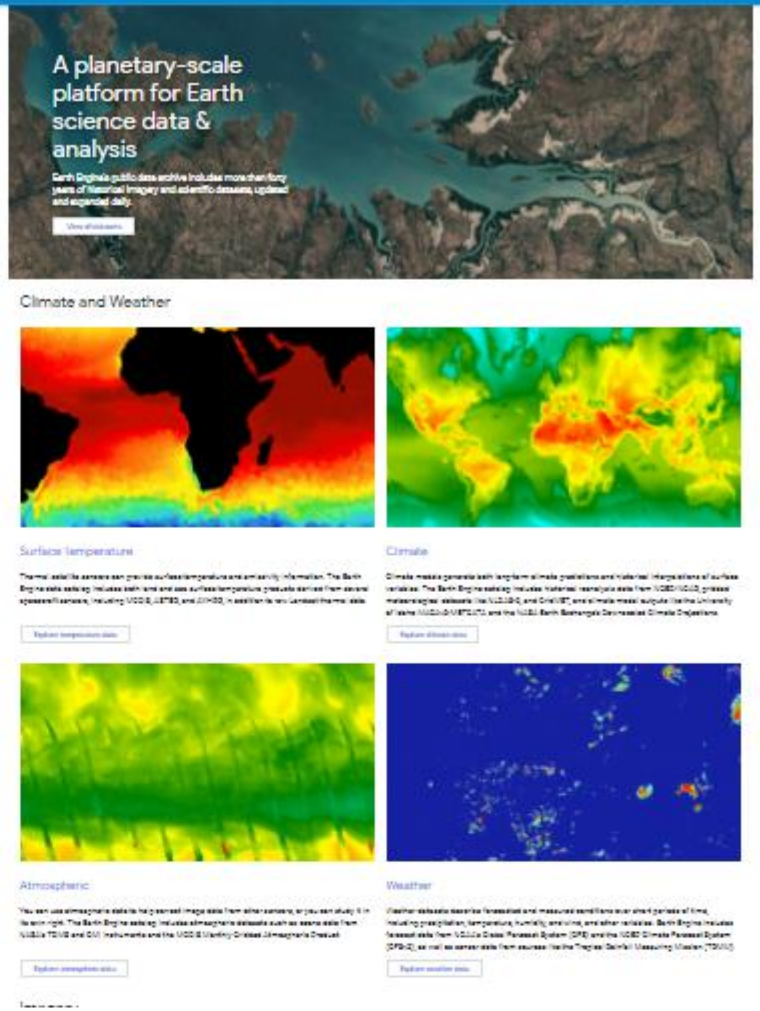
Data Catalog



Data catalog contains a description of the available datasets by theme

https://developers.google.com/earth-engine/datasets/

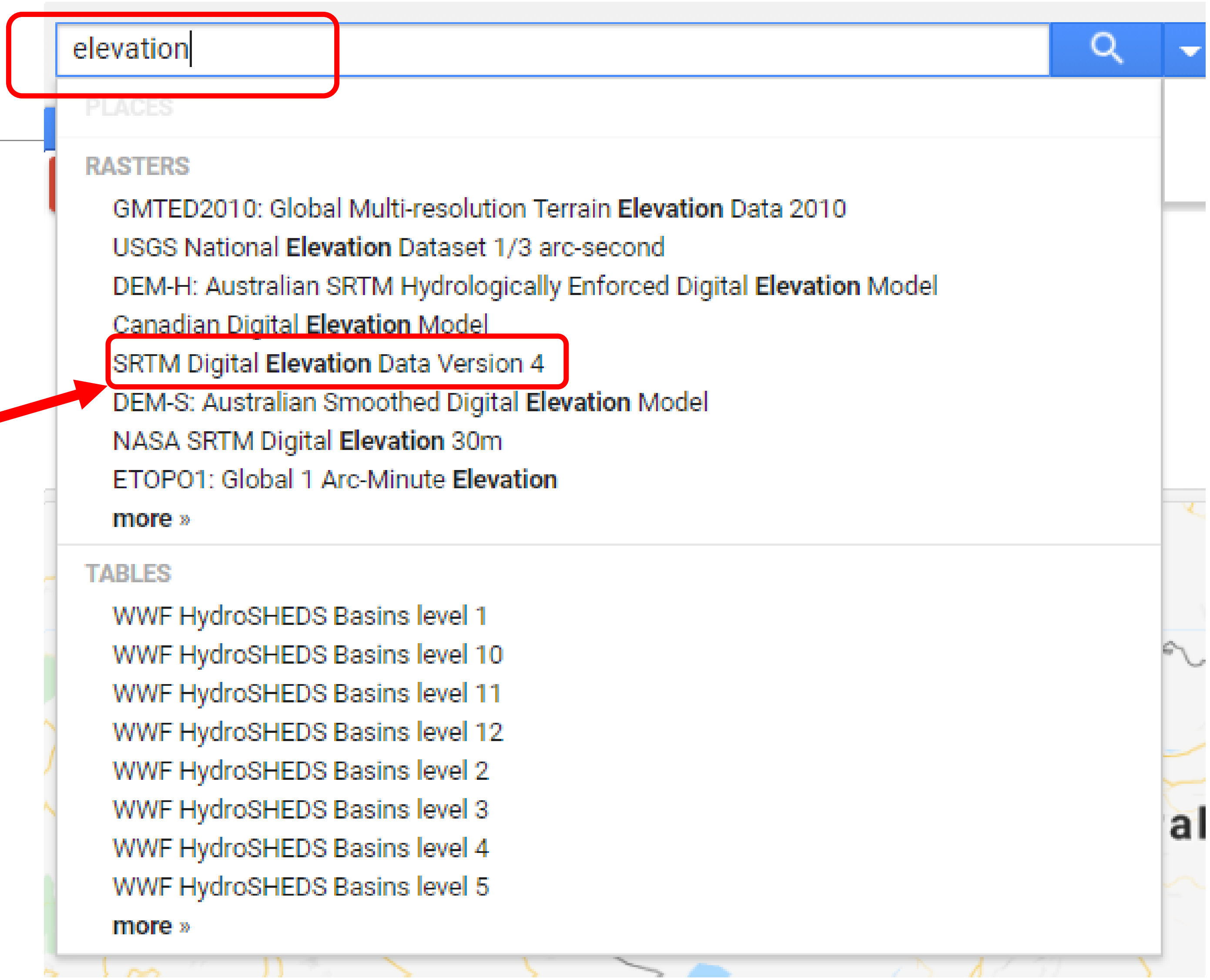
Search or directly select data from the catalog to import



Search Data - Elevation

Write key words in search window.

Select data



SRTM Digital Elevation Data Version 4

DESCRIPTION BANDS TERMS OF USE CITATIONS

The Shuttle Radar Topography Mission (SRTM) digital elevation dataset was originally produced to provide consistent, high-quality elevation data at near global scope. This version of the SRTM digital elevation data has been processed to fill data voids, and to facilitate its ease of use.

Dataset Availability

2000-02-11T00:00:00 - 2000-02-22T00:00:00

Dataset Provider

NASA/CGIAR

Collection Snippet

```
ee.Image("CGIAR/SRTM90_V4")
```

Tags

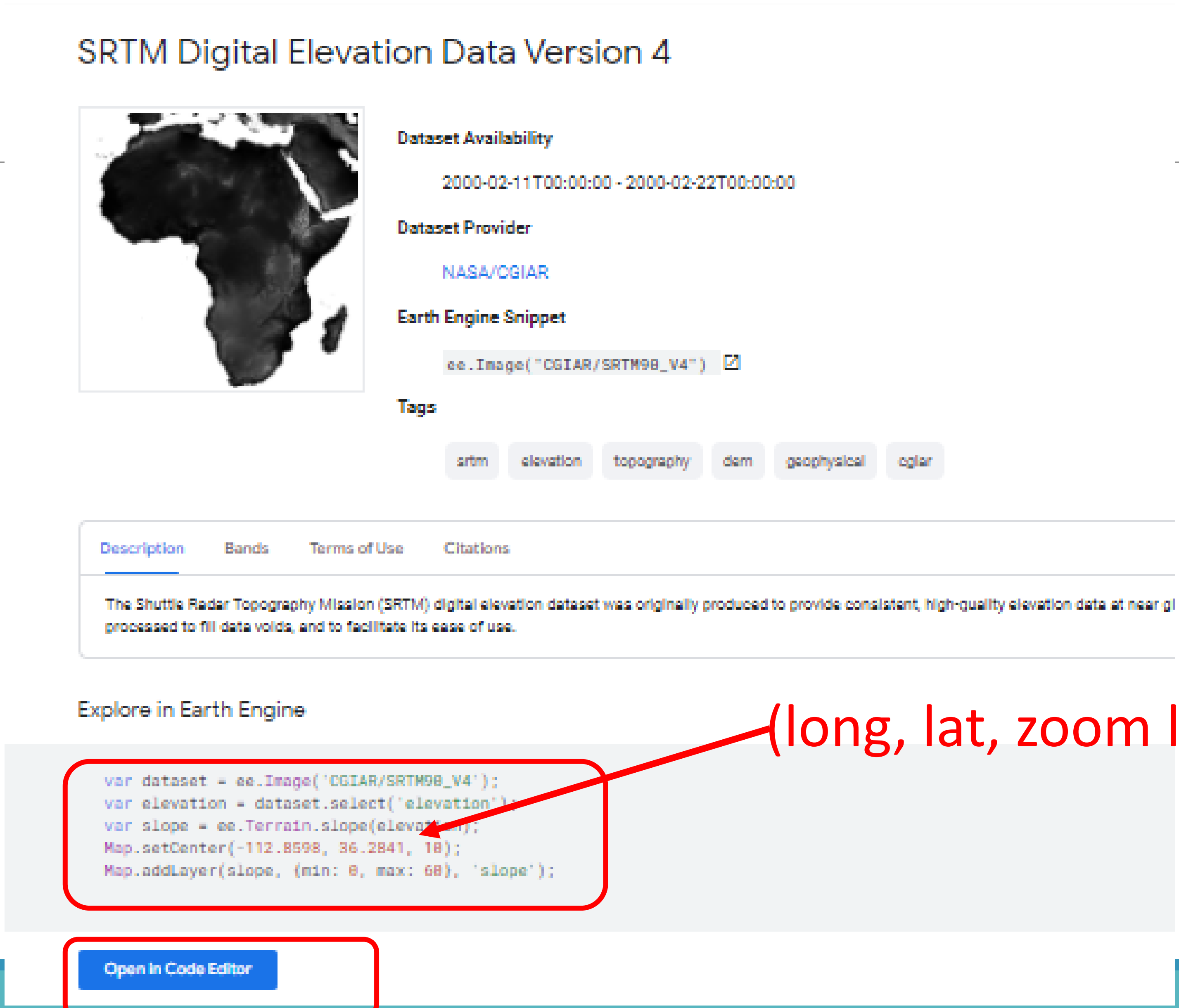
srtm elevation topography

dem geophysical cgiar

Or- View Dataset in Catalog Page

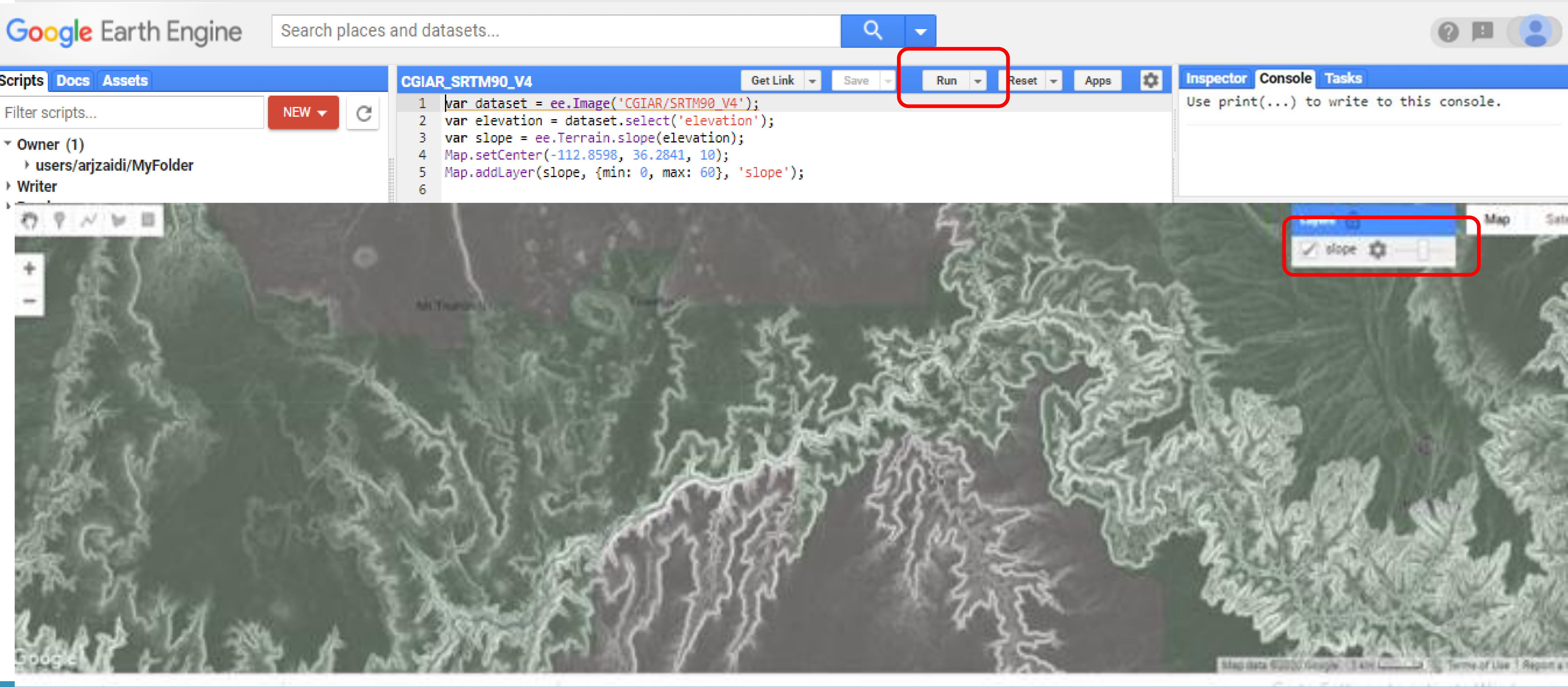
CLOSE IMPORT

View Dataset in Catalog Page



(long, lat, zoom level)

Open Code Editor



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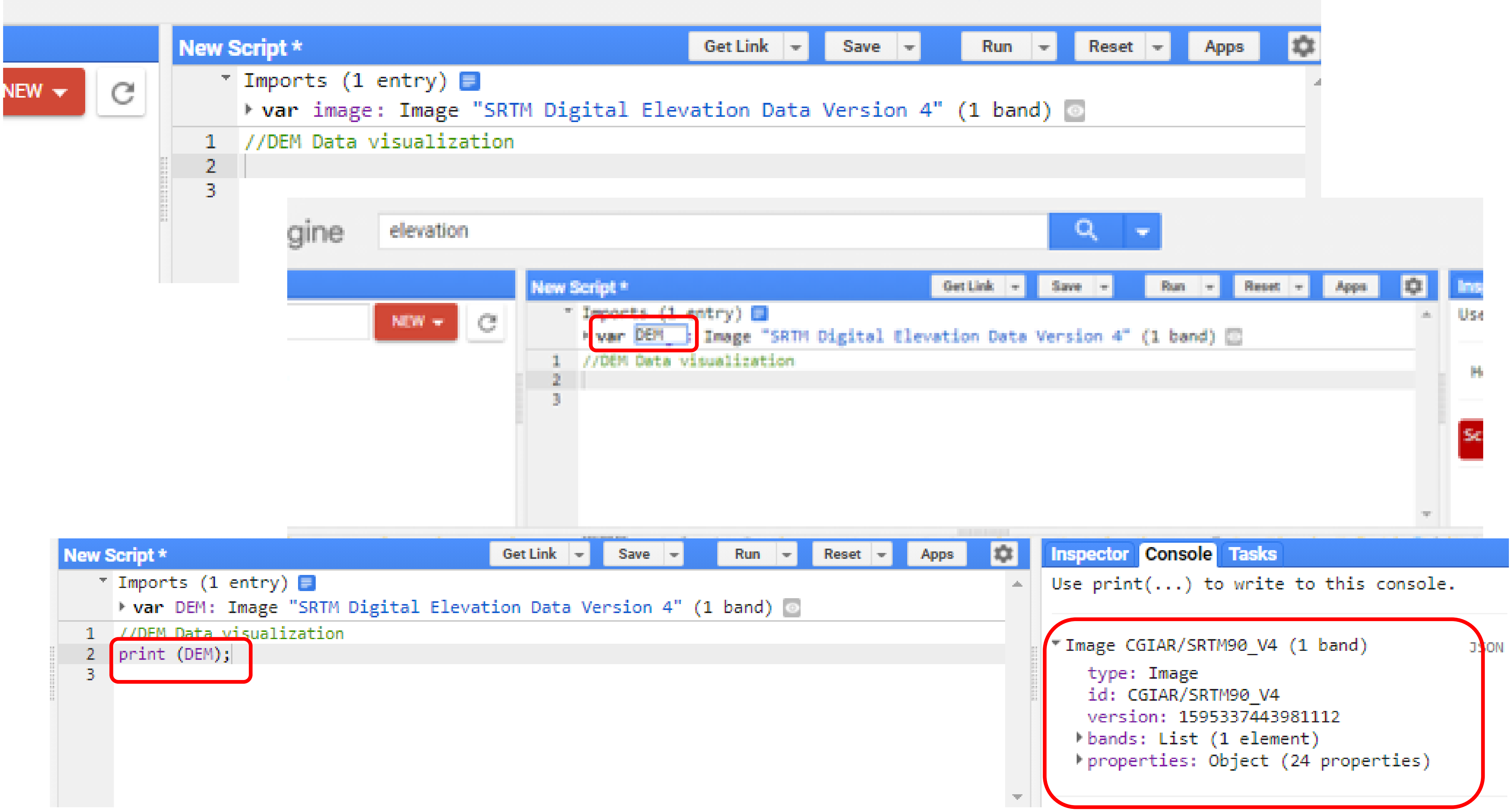
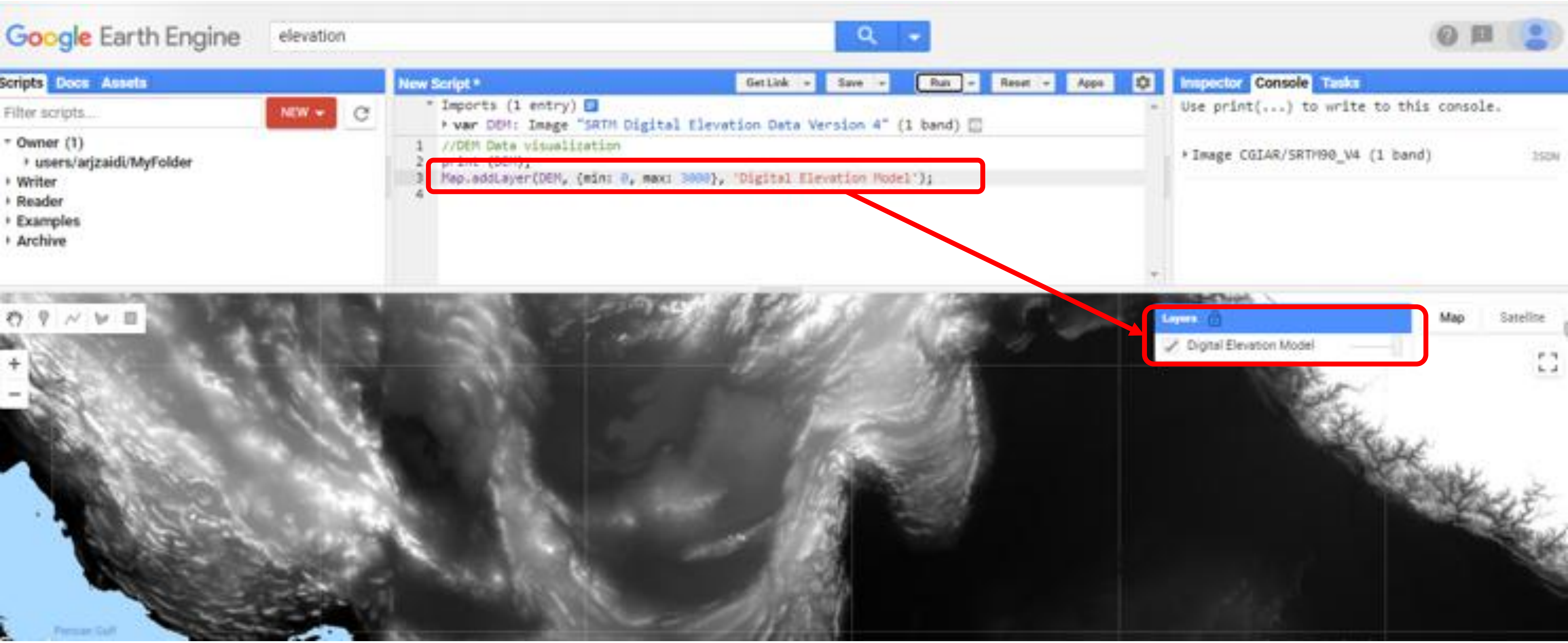
```
ee.Image("CGIAR/SRTM90_V4")
```

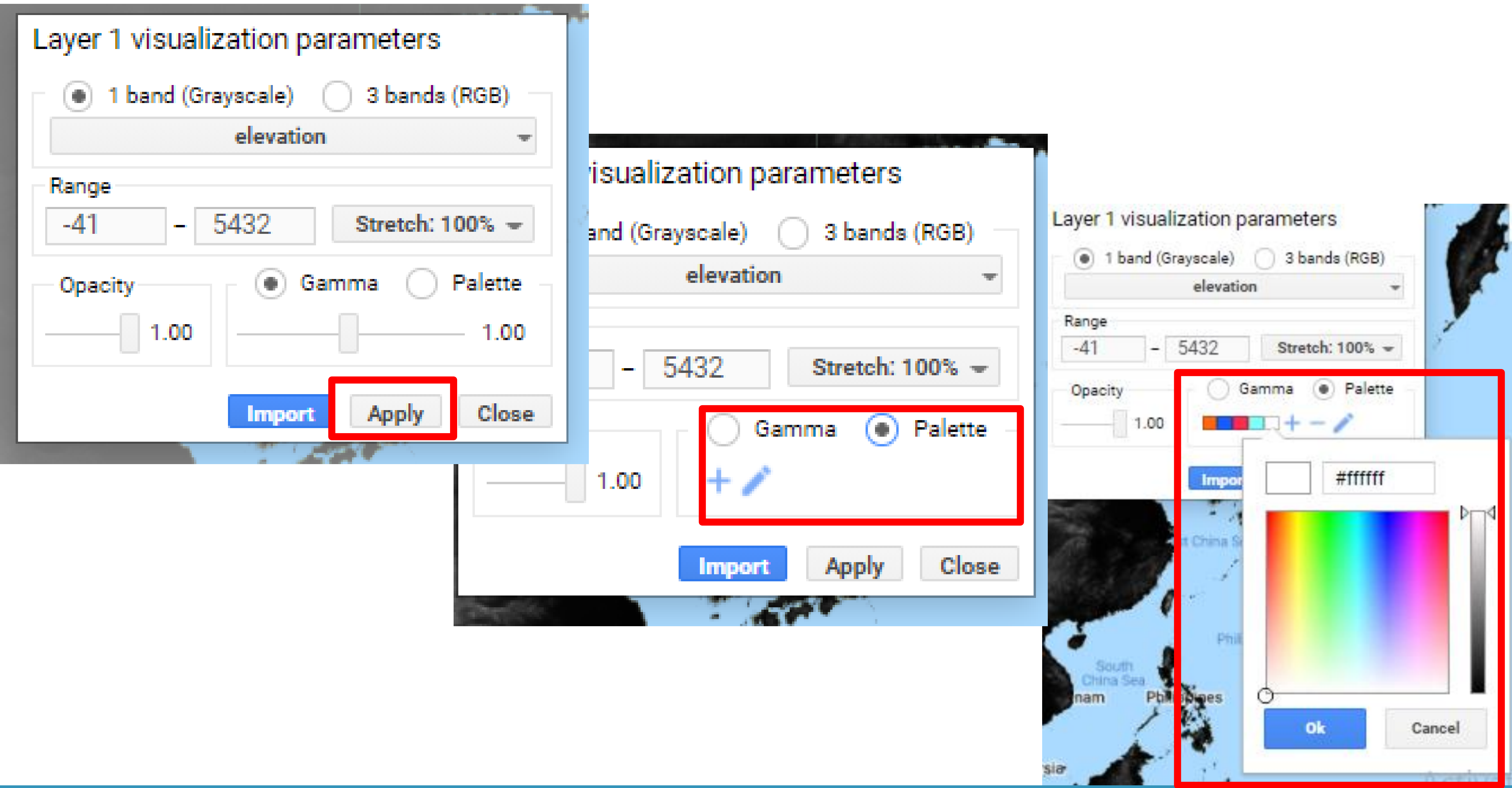
Tags

srtm elevation topography dem geophysical cgiar

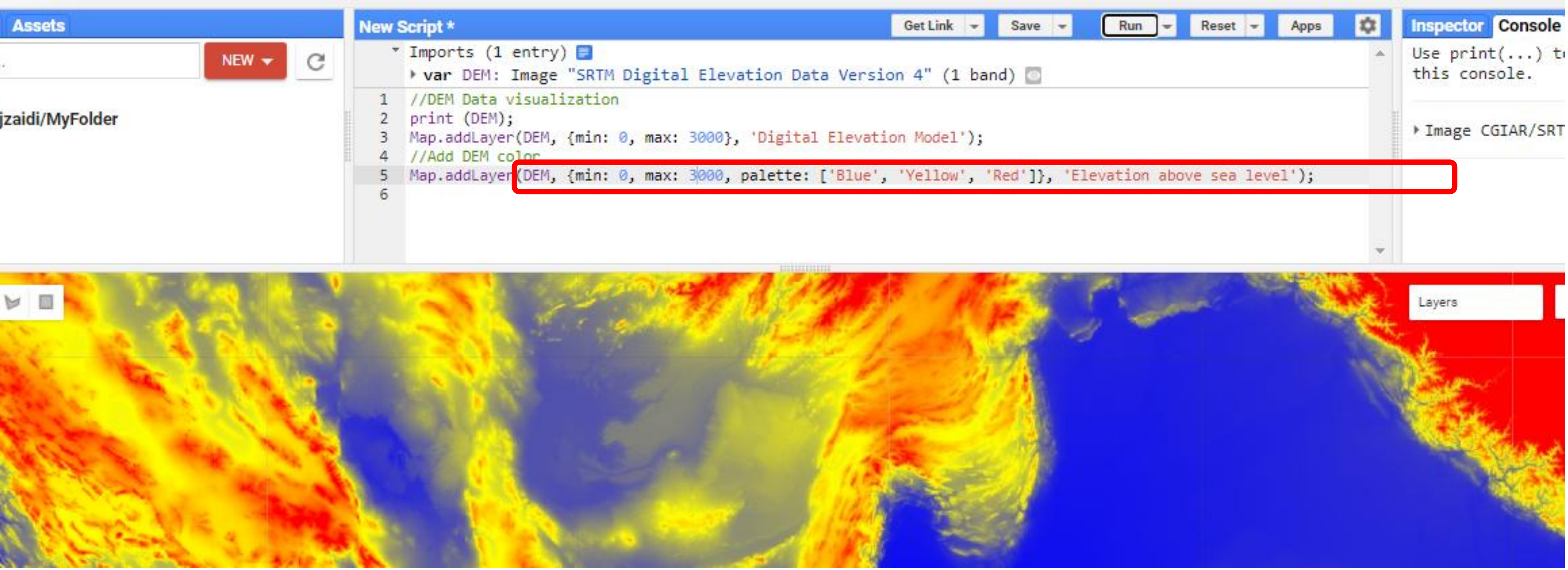
CLOSE IMPORT


New Script *
Get Link
Save
Run
Reset
Apps
NEW
Imports (1 entry)
var image: Image "SRTM Digital Elevation Data Version 4" (1 band)
//DEM Data visualization
elevation
var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band)
print (DEM);
Inspector
Console
Tasks
Use print(...) to write to this console.
Image CGIAR/SRTM90_V4 (1 band)
JSON
type: Image
id: CGIAR/SRTM90_V4
version: 1595337443981112
bands: List (1 element)
properties: Object (24 properties)

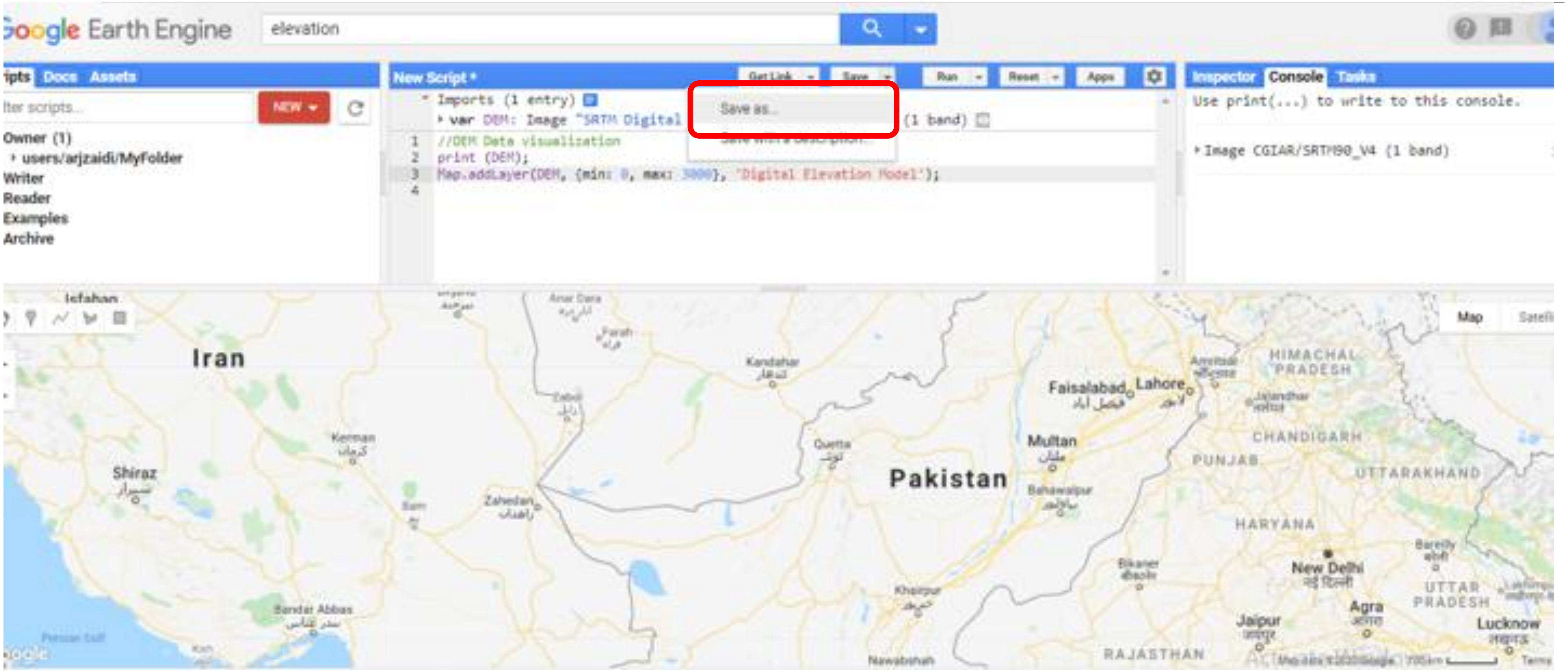

Google Earth Engine
elevation
Scripts
Docs
Assets
Filter scripts...
NEW
Owner (1)
users/arjzaidi/MyFolder
Writer
Reader
Examples
Archive
New Script *
Get Link
Save
Run
Reset
Apps
Imports (1 entry)
var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band)
//DEM Data visualization
Map.addLayer(DEM, {min: 0, max: 3000}, 'Digital Elevation Model');
Inspector
Console
Tasks
Use print(...) to write to this console.
Image CGIAR/SRTM90_V4 (1 band)
Layers
Digital Elevation Model
Map
Satellite


Layer 1 visualization parameters
1 band (Grayscale)
3 bands (RGB)
elevation
Range
-41
5432
Stretch: 100%
Opacity
Gamma
Palette
1.00
1.00
Import
Apply
Close
isualization parameters
and (Grayscale)
3 bands (RGB)
elevation
5432
Stretch: 100%
Gamma
Palette
1.00
Import
Apply
Close
Layer 1 visualization parameters
1 band (Grayscale)
3 bands (RGB)
elevation
Range
-41
5432
Stretch: 100%
Opacity
Gamma
Palette
1.00
#ffffff
South China Sea
Ok
Cancel

Change Symbology Color



Saving your Script



THANK YOU